

Priya Ranjan

Computer Science Undergraduate • Software & Systems Builder

"I build thoughtful software experiences at the intersection of technology, problem solving and intelligent systems."

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🎓 EDUCATION

Bachelor of Technology (B.Tech) in Computer Science & Engineering Specialization in Artificial Intelligence & Distributed Systems 2023 – 2027

Core Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems, Theory of Computation, Discrete Mathematics, Object-Oriented Software Design.

</> TECHNICAL SKILLS

Core Languages: Python, TypeScript, JavaScript (ES6+), C, SQL, HTML5 / Modern CSS

Frameworks & Engines: Next.js 15 (App Router), React 18, FastAPI, Node.js / Express, Flask, TailwindCSS / CSS Modules

AI & Data Science: PyTorch, OpenCV, Scikit-Learn, XGBoost, SHAP (Explainable AI), NumPy & Pandas

Cybersecurity & Networks: Nmap & Port Auditing, OWASP ZAP / Web Security, STRIDE Threat Modeling, Wireshark & Packet Analysis, Linux Socket Scripting

Systems & Infrastructure: OS Disk Scheduling Algorithms, PostgreSQL, MongoDB & Geospatial (2dsphere), Redis, Docker & Compose, Git & GitHub Collaboration

VIREONIQ — Developer Career Roadmap & Code Analysis Engine

2024

A developer skill assessment platform that parses submitted code using Python Abstract Syntax Trees (AST) and generates clear, step-by-step learning roadmaps instead of relying on keyword-stuffed resumes.

- **Problem & Architecture:** Traditional hiring platforms rely on keyword search. They reward candidates who stuff buzzwords into resumes, but cannot tell if someone understands foundational data structures, asynchronous event loops, or clean system design.
- **Engineered Result:** Built an asynchronous FastAPI backend paired with a React frontend. It parses code submissions into an Abstract Syntax Tree to compute structural complexity metrics without execution risk, compares the profile against job market vectors in Qdrant, and returns an ordered prerequisite roadmap.

Technologies: Python, FastAPI, React 18, PyTorch, Qdrant Vector DB, PostgreSQL, Redis, Docker Compose

TRUSTSHIELD X — Concurrent Threat Scanner & Security Analysis Tool

2024

A fast, concurrent security scanner built in Python 3.13 asyncio that tests open ports, security headers, and payment links in under 15ms with tamper-evident audit logs.

- **Problem & Architecture:** Developers and small teams frequently use three or four disparate tools to inspect network ports, HTTP header configurations, and suspicious URLs. Most scanners are slow, sequential, and produce messy unstructured outputs.
- **Engineered Result:** Engineered a concurrent scanner in Python 3.13 using asyncio and FastAPI. Probes network ports, checks HTTP security headers (CSP, HSTS, X-Frame-Options), and validates payment links concurrently within 15ms. Emits SHA-256 hashed audit records for tamper verification.

Technologies: Python 3.13, FastAPI, AsyncIO, PyCryptodome, PostgreSQL, Docker

TECH ON TOUR — Intelligent Regional Exploration & Tourism Graph Engine

2024

A location-intelligent travel and regional discovery platform utilizing graph route traversal and spatial datasets to plan cultural heritage itineraries.

- **Problem & Architecture:** Travelers exploring heritage and regional tourism circuits struggle with scattered information, unreliable opening schedules, and fragmented transport links, resulting in inefficient transit and missed cultural landmarks.
- **Engineered Result:** Developed a full-stack platform combining structured CSV and GeoJSON spatial datasets with graph route traversal algorithms. Provides interactive map views, transit details, and historical context for cultural monuments.

Technologies: Python, Pandas, Flask / FastAPI, Leaflet / Mapbox, HTML5 Canvas, Vanilla JS, CSV / GeoJSON

An intelligent career platform combining spaCy natural language entity extraction, ATS compatibility grading, and Random Forest machine learning classification.

- **Problem & Architecture:** Applicant Tracking Systems (ATS) reject qualified resumes due to non-standard formatting or missing arbitrary keywords, leaving job seekers without actionable feedback on why their application was disqualified.
- **Engineered Result:** Engineered an end-to-end pipeline with React 19 and FastAPI. Ingests PDF resumes, runs spaCy Named Entity Recognition to identify technical tools and credentials, scores ATS formatting compatibility, and runs a Random Forest classifier for candidate-role matching.

Technologies: Python 3.12+, FastAPI, React 19, Vite, Tailwind CSS v4, spaCy NLP, Scikit-Learn, PyPDF2

PRIOCARDIX AI — Cardiovascular Risk Calculator & Scenario Simulator

2024

A responsive healthcare web app built with React and Zustand that calculates cardiovascular risk tiers and lets users simulate how lifestyle improvements lower their risk in real time.

- **Problem & Architecture:** Traditional online health risk calculators are opaque and discouraging. They output an arbitrary percentage score without explaining which biomarkers contributed most or how specific lifestyle adjustments would change the outcome.
- **Engineered Result:** Engineered a clean React and Zustand application that calculates clinical risk tiers from blood pressure, resting heart rate, lipid levels, and BMI. Included interactive scenario sliders that recompute risk in real time without lag.

Technologies: React, Vite, Tailwind CSS, Zustand, Chart.js

🏆 FOUNDATIONS & ACHIEVEMENTS

- **Operating System Kinematics Research:** Built open-source Disk Scheduling Algorithm simulator comparing FCFS, SSTF, SCAN, and C-LOOK head traversals to evaluate seek latency limits.
- **Algorithmic Problem Solving:** Regular practice across competitive programming platforms, focusing on graph algorithms, dynamic programming, and asymptotic time complexity optimizations.
- **Explainable Clinical AI:** Integrated TreeSHAP game-theoretic attribution with cardiovascular gradient-boosted ensembles in Priocardix AI to ensure clinical transparency.